Structural Self-Energy Expansion (SSEE-V3.6): A Minimal-Parameter Dark Energy Framework — Algebraic Derivation, Boltzmann Validation, and Effective Field Theory Stability Analysis

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Abstract

We present the Structural Self-Energy Expansion (SSEE-V3.6), a dark energy framework that derives its cosmological background sector algebraically from the golden ratio φ and π, with zero fitted dimensionless parameters at the background level and no WIMP, QCD-axion, or SUSY dark-matter particle. Phenomenological extensions in the dark-matter sector (Paper 6) are currently treated as ansätze and tracked as open problems (OP-9/11, `OPEN_PROBLEMS.md`). In CMB-likelihood terms SSEE is a $k = 2$ fit: of ΛCDM's six parameters it fixes four algebraically (Ω_b, Ω_c via $\omega_c = \mathrm{KAL}_0\, \omega_b\, n_s$, $n_s = 1 - \varphi^{-7}$, and the derived anchor $H_0 = 3(\varphi + \pi)^2$), leaving only the primordial amplitude A_s and the optical depth τ ($k = 2$ versus $k = 6$, the count used in the Paper 3 model comparison). The dark energy equation of state $w_0 = -(3\varphi + \pi)/[2(\varphi + \pi)] \simeq -0.840$ and $w_a \simeq -0.670$ are fixed by the algebraic construction and agree with the official DESI DR2 w_0w_aCDM posterior within 0.24σ (Pantheon+; 0.2–1.8σ across SN compilations)—a parameter-free postdiction, with no claim of temporal priority over DESI DR2. A Bayesian MCMC analysis under the ω_m-direct CMB anchor as prior ($H_0^{\mathrm{prior,SSEE}} = 67.962 \pm 0.54$ km s^{-1} Mpc^{-1}, the value at which Planck `plik_lite` is minimised with SSEE-fixed ω_b, ω_c; Paper 3) yields $H_0 = 67.95^{+0.40}_{-0.40}$ km s^{-1} Mpc^{-1} (0.88σ from Planck, 0.04σ from the algebraic anchor) and ΔBIC $= 5.68$ favouring SSEE over ΛCDM (5.59 over CPL). A complementary full-likelihood multi-probe MCMC (CAMB + Planck `plik_lite` + lensing + DESI DR2 + $f\sigma_8$, w_0, w_a free and H_0 under a flat prior) places the model at 1.31σ in the w_0–w_a plane, where ΛCDM is excluded at 3.22σ; under this uncompressed CMB likelihood H_0 relaxes to 65.87 ± 1.17 (1.79σ below the anchor).

The CMB-sector matter density is the standard physical observable $\Omega_{m,\mathrm{CMB}} = \omega_m/h^2 = 0.30889$ (Planck: 0.3153 ± 0.0073), built piece-by-piece from SSEE algebra (ω_m-direct; OP-8 dissolved, no matter factor) and distinct from the dynamical $\Omega_{m,\mathrm{dyn}} = 0.160$ that governs DESI BAO. Full-sky CLASS Boltzmann runs confirm that the *full* matter density is physically necessary: using only $\Omega_{m,\mathrm{dyn}} = 0.160$ as the CMB matter density shifts the peak positions by $\sim 10\%$ and raises the RMS residual from 1.4% to 31.5%.

A linear Israel-Stewart perturbation analysis with $\Omega_{m,\mathrm{cosm}} = 0.30889$ in the Poisson equation yields a growth factor $\mathcal{G} = 1.0032$ and $\sigma_8 = 0.8136$ (Paper 5), giving a mean $f\sigma_8$ tension of 0.70σ across six RSD surveys — statistically identical to ΛCDM (0.73σ), resolving the growth-rate sector. A two-sector ϕ-DM extension — $\Omega_{\phi\mathrm{DM}} = \Omega_{m,\mathrm{CMB}} - \Omega_{m,\mathrm{dyn}} = 0.14889$ active only on scales $k < k_{\mathrm{fs}} = 0.754\, h\,\mathrm{Mpc}^{-1}$ — reduces the S_8 lensing tension from 3.5σ (single-sector ceiling, KiDS) to 0.04σ, with $\sigma_8^{\mathrm{eff}} = 0.747$ ($S_8^{\mathrm{eff}} = 0.758$ from the direct two-sector CLASS run; the analytic two-sector growth factor $G_{2\mathrm{s}} = 1.003$ is ΛCDM-consistent, the suppression to $\sigma_8^{\mathrm{eff}} = 0.747$ arising from ϕ-DM free-streaming below k_{fs}). The same suppression reaches the $\sigma_8(R{=}8)$ window probed by RSD, so the two-sector $f\sigma_8$ tension rises modestly from the 0.70σ single-sector baseline to 0.93σ (still $< 1\sigma$, close to ΛCDM 0.73σ) — the deliberate cost of lowering σ_8 to resolve the lensing amplitude.

An EFTCAMB validation at the Bellini-Sawicki level uses the fluid-approximation input $\alpha_K^{\mathrm{fluid}} = 0.4033$ ($\Delta = 0.005\%$), while the theoretical bare-field value is $\alpha_K = 0.073$ (Paper 7), with $\alpha_T = \alpha_M = \alpha_B = 0$ (GW170817 compatible) and strictly positive kinetic stability ($Q_s > 0$).

We additionally summarise three theoretical extensions of the framework. In the strong-field regime Paper 8 analyses two limits: an alternative MOND-like limit in which the photon deflection is amplified by $\sqrt{\mathrm{AURA}} \approx 1.9995$ (numerically close to $\mathcal{M}$ at 0.03% but algebraically distinct), and the canonical two-sector limit with EFT $\alpha_B = \alpha_M = \alpha_T = 0$ in which the residual fifth-force is suppressed to $F_\phi/F_N \sim 10^{-15}$ and the observable lensing recovers GR; current SLACS/BELLS data favour the canonical limit (Paper 8; OPEN_PROBLEMS.md OP-13). A late-time local screening fraction $f_{\mathrm{scr}} = \alpha_K/(3\mathcal{M}) = (\pi - \varphi)/(\varphi + \pi)^2 \simeq 0.067$ applied to the global background $H_0^{\mathrm{alg}} = 3(\varphi + \pi)^2 = 67.962$ gives $H_0^{\mathrm{local}} = 72.86\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$, 0.17σ from the SH0ES distance ladder, as a zero-parameter prediction of the infrared sector (Paper 9). A UV completion $K(X) = X/\mathrm{KAL}_0 + X^2/M^4$ with algebraic cutoff $M = 9.68$ meV connects the dark-energy EFT to the inflationary α-attractor (Paper 10). All model parameters are falsifiable by named forthcoming experiments.

Contents

1 Introduction

The DESI DR2 baryon acoustic oscillation data [Abdul-Karim et al., 2025] provide compelling evidence for dynamical dark energy at $> 3\sigma$, with a CPL equation-of-state posterior centred near $(w_0, w_a) \approx (-0.84, -0.67)$. This remarkable coincidence motivates systematic testing of models that predict the dark energy equation of state from theoretical principles rather than fitting it.

The SSEE-V3.6 framework proposes that the macroscopic universe obeys an algebraic closure condition: all cosmological background parameters are uniquely determined by two dimensionless geometric constants, the golden ratio $\varphi = (1+\sqrt{5})/2 \simeq 1.6180$ and $\pi \simeq 3.1416$. No optimisation over formula space was performed: the parameter values are fixed by the algebraic construction [Almeida Vallejo, 2026k], not adjusted to the data. We make no claim of temporal priority over DESI DR2 (public from 2025 March); the agreement reported below is a parameter-free *postdiction*, and the only timestamped predictions are those reserved for unreleased data (DESI DR3, Euclid).

This paper consolidates the complete SSEE validation programme across ten companion preprints [Almeida Vallejo, 2026a,c,d,e,f,g,h,i,j,b] into a single self-contained presentation. We report four independent validation tiers:

1. **Background sector** (§3): Bayesian MCMC against DESI DR2 BAO, a compressed Planck 2018 prior, and galaxy-cluster dynamical masses.
2. **CMB angular spectrum** (§4): Full-sky CLASS Boltzmann integration demonstrating that the full ω_m-direct matter density ($\Omega_{m,\mathrm{CMB}} = 0.30889$) is physically necessary for reproducing Planck PR4 peak positions.
3. **Structure growth and σ_8** (§5): Israel-Stewart causal perturbation theory and a ϕ-DM two-sector extension that resolves the S_8 lensing tension via free-streaming suppression (at a modest $\sim 0.2\sigma$ cost in $f\sigma_8$, which rises to 0.93σ).
4. **EFT stability** (§6): EFTCAMB Bellini-Sawicki validation of kinetic stability (α_K) and gravitational wave compatibility ($\alpha_T = \alpha_M = \alpha_B = 0$).

Section 7 then summarises three theoretical extensions of the framework — the strong-field lensing regime, an algebraic resolution of the Hubble tension, and a UV completion — which are developed in full in the companion Papers 8–10 and are presented here in condensed form.

Throughout, we distinguish three epistemic statuses: *postdiction* (algebraic result fixed by construction, then compared to data already public), *retrodiction* (independently derived, then compared to existing data), and *prediction* (fixed now, tested against not-yet-released data). Table 6 in §8 lists all entries explicitly.

2 Algebraic Framework

2.1 Structural registers

The SSEE framework postulates a closed algebraic system of seven structural registers derived from φ and π:

$$\begin{aligned}
\Omega &= \pi + \varphi \simeq 4.7596 \quad \text{(Stability Metric)}\\
\beta &= (\pi + \varphi)/2 \simeq 2.3798 \quad \text{(Base Coupling Scalar)}\\
\mathrm{KAL}_0 &= \beta + \pi \simeq 5.5214 \quad \text{(Structural Viscosity)}\\
P_{\mathrm{sc}} &= \Omega + \varphi \simeq 6.3776 \quad \text{(Dynamical Evolution Scalar)}\\
K_v &= \varphi + \pi + \Omega \simeq 9.5192 \quad \text{(Structural Constraint)}\\
T_r &= 3(\varphi + \beta) \simeq 11.9935 \quad \text{(3D Saturation Horizon)}\\
M_v &= \varphi + \pi + K_v \simeq 14.2788 \quad \text{(Maximal Dimensional Invariant)}
\end{aligned} \tag{1}$$

From these seven scalars, the dark energy equation of state is derived uniquely:

$$w_0 = -\frac{T_r}{M_v} = -\frac{3\varphi + \pi}{2(\varphi + \pi)} \simeq -0.8399, \qquad w_a = -\frac{P_{\mathrm{sc}}}{K_v} = -\frac{2\varphi + \pi}{2(\varphi + \pi)} \simeq -0.6699. \tag{2}$$

The dark energy density fraction follows algebraically: $\Omega_{\mathrm{DE}} = |w_0| = T_r/M_v \simeq 0.8399$.

2.2 Derived cosmological parameters

All primary observational inputs are retrodicted without fitting:

$$H_0 = 3(\varphi + \pi)^2 \simeq 67.96\ \mathrm{km\,s^{-1}\,Mpc^{-1}}, \tag{3}$$

$$n_s = 1 - \varphi^{-7} \quad \simeq 0.96556, \tag{4}$$

$$\Omega_{m,\mathrm{CMB}} = \omega_m/h^2 \quad \simeq 0.30889 \quad (\omega_m\text{-direct, CMB sector}). \tag{5}$$

Planck 2018 measurements: $H_0 = 67.36 \pm 0.54$, $n_s = 0.9649 \pm 0.0042$, $\Omega_m = 0.3153 \pm 0.0073$ [Planck Collaboration, 2020]. All three SSEE retrodictions lie within 1.5σ of the Planck values with no adjustment.

2.3 The two-Ω_m structure and CMB matter density

The framework predicts two independent matter densities. The dark energy EoS predicts $\Omega_{\mathrm{DE}} = |w_0| = 0.840$, which implies a *dynamical* matter density $\Omega_{m,\mathrm{dyn}} = 1 - \Omega_{\mathrm{DE}} = 0.160$. However, the CMB angular-diameter distance is sensitive to the *total* matter density, which under the ω_m-direct reframe (OP-8 dissolved) is the standard physical observable built from SSEE algebra:

$$\Omega_{m,\mathrm{CMB}} = \frac{\omega_m}{h^2} = \frac{\omega_b + \omega_c + \omega_\nu}{h^2} = 0.30889, \qquad \omega_c = \mathrm{KAL}_0\, \omega_b\, n_s, \tag{6}$$

in agreement with Planck 2018 (0.3153 ± 0.0073) at 0.88σ. There is *no* matter factor: $\Omega_{m,\mathrm{dyn}} = 0.160$ (DESI) and ω_m (CMB) are two independent SSEE predictions. (The earlier, now superseded MIRA mapping $\mathcal{M} = (3\varphi + \pi)/4 \simeq 1.9989$ gave $\Omega_{m,\mathrm{CMB}} = \mathcal{M} \times \Omega_{m,\mathrm{dyn}} = 0.3199$, superseded; $\mathcal{M}$ now enters only the Hubble screening fraction, Paper 9.) The physical interpretation of this two-sector split — whether it arises from a background-level Israel-Stewart thermodynamic correction, a two-sector dark matter architecture, or both — is examined in Papers 3 and 5 [Almeida Vallejo, 2026d,f]. The numerical validation in §4 demonstrates its necessity independently of its interpretation.

3 Background Sector Validation

3.1 Bayesian MCMC setup

We compare SSEE against ΛCDM and a free CPL parametrisation using Bayesian evidence criteria and the EMCEE ensemble sampler [Foreman-Mackey et al., 2013] with $N_{\mathrm{walkers}} = 100$, $N_{\mathrm{steps}} = 25\,000$ (2.5×10^6 total evaluations). The likelihood combines:

- DESI DR2 BAO: 13 data points at $z = 0.295$–2.330 [Abdul-Karim et al., 2025], with the full 13×13 same-bin block-diagonal covariance;
- Planck 2018 compressed Gaussian prior: $H_0 = 67.36 \pm 0.54$, $\Omega_m = 0.3153 \pm 0.0073$, $\Omega_b h^2 = 0.02237 \pm 0.00015$, $\rho(H_0, \Omega_m) = -0.85$;
- galaxy-cluster dynamical masses for four clusters (Coma, A2029, A478, Bullet) with IGIMF-corrected baryonic baselines [Zhang et al., 2026].

For SSEE, H_0 is the sole free parameter; (w_0, w_a) are fixed algebraically. For ΛCDM, (H_0, Ω_m) are free. For CPL, $(H_0, \Omega_m, w_0, w_a)$ are free.

3.2 Results

Table 1 summarises the key posteriors and information criteria.

Table 1: Bayesian MCMC results (three-model run, DESI DR2 + common Planck prior, total matter density in the background geometry). N_{eff}: effective independent samples. Positive Δ favours SSEE.

Quantity	SSEE	ΛCDM	CPL
H_0 [km s^{-1} Mpc^{-1}]	67.62 ± 0.40	68.27 ± 0.28	67.26 ± 0.51
Free parameters k	2	3	5
N_{eff}	78 750	64 505	44 627
ΔBIC relative to SSEE	ref	+5.68	+5.59
ΔDIC relative to SSEE	ref	+4.91	+3.27
χ_r^2 clusters	0.126	0.081	—
(w_0, w_a) vs DESI DR2 $w_0 w_a$CDM (Pantheon+)	0.24σ	0.7σ	—

With the total gravitating matter density $\Omega_m = \omega_m/h^2 = 0.30889$ in the background geometry, SSEE is favoured on every information criterion: ΔBIC = 5.68 over ΛCDM and 5.59 over CPL, ΔDIC = 4.91 and 3.27. The SSEE sound horizon then matches ΛCDM (r_d = 148.2 Mpc vs 147.8 Mpc) and the H_0 posterior sits 0.39σ from Planck, so no background-structure penalty arises. An earlier "+206" full-background penalty was an artefact of placing the cold sector $1 + w_0 = 0.160$ in $E(z)$ where the total density belongs; with the correct total density it does not occur. The ω_m-direct two-Ω_m construction (§2.3) resolves the background tension, validated in §4 below.

3.3 Galaxy-cluster mass sector

The four-cluster analytic forward model — baryon fraction $f_b = \Omega_b/\Omega_{m,\mathrm{CMB}}$ corrected by IGIMF baryonic masses [Zhang et al., 2026] — achieves $\chi_r^2 = 0.122$ (four-cluster MCMC subset) and $\chi^2_r = 0.126$ (seven-cluster analytic). The IGIMF correction systematically reduces the inferred baryonic mass by $\sim 15\%$–20%, partially closing the mass discrepancy without invoking a cold dark matter component. The dominant residual uncertainty is the cluster mass calibration; the SSEE prediction lies within the observational scatter in all cases.

3.4 Multi-probe MCMC validation: two complementary tests

The rigidly fixed SSEE background can be interrogated in two distinct and complementary ways, which must not be conflated. *(i) Free test:* let the data choose the equation of state and ask where they land relative to the algebraic prediction — *"what do the data prefer?" (ii) Fixed test:* impose the algebraic values and measure the absolute goodness of fit — *"how well does the imposed model describe the data?"* Both are performed with the *full* CMB power-spectrum likelihood — not a compressed distance prior — so that no ΛCDM-like assumption is smuggled in through the compression.

Free test (what the data prefer). We run an MCMC with CAMB + Planck `plik_lite` TTTEEE + lensing + a τ prior, combined with DESI DR2 BAO and $f\sigma_8$, sampling $(H_0, \Omega_b h^2, \Omega_c h^2, n_s, \tau, w_0, w_a)$ with *flat* priors (H_0 free in $[60, 80]$; four chains, $\sim$53k post-burn-in samples, Gelman–Rubin $R - 1 < 0.005$). Table 2 lists the recovered posteriors: the tightly constrained CMB combinations $\omega_b, \omega_c, r_{\rm drag}$ are recovered to $\leq 0.2\sigma$ — an independent confirmation of the ω_m-direct reframe from the full C_ℓ spectrum rather than from a compressed prior. The remaining parameters $(H_0, \Omega_m, w_0, w_a)$ sit 1.4–1.8σ from the algebraic values, the uncompressed CMB likelihood pulling H_0 down to 65.87 ± 1.17. In the dark-energy plane the free posterior $(w_0, w_a) = (-0.655, -1.104)$ *excludes* the cosmological constant $(-1, 0)$ at 3.22σ, while the SSEE algebraic point $(-0.840, -0.670)$ lies at only 1.31σ in the correlated ($\rho = -0.93$) w_0–w_a plane — more than twice as close as ΛCDM.[1]

Under the *full* Planck C_ℓ likelihood — with H_0 free under a flat prior rather than fixed to a compressed distance prior — the posterior sits at $H_0 = 65.87 \pm 1.17$, some 1.79σ below the algebraic anchor $H_0^{\rm alg} = 67.96$. This mild downward pull is a known feature of $w_0 w_a$CDM under the uncompressed CMB likelihood and is stated openly. The key model-comparison statement is unchanged and in fact sharper: the fixed algebraic point $(-0.840, -0.670)$ lies 1.31σ from the joint w_0–w_a posterior, whereas the cosmological-constant point $(-1, 0)$ is excluded at 3.22σ. The joint distance is smaller than the individual w_0, w_a marginals (1.73σ, 1.43σ) because the algebraic offset lies *along* the strong w_0–w_a degeneracy ($\rho = -0.93$), not across it. Corner plots and trace diagnostics are archived at Zenodo [Almeida Vallejo, 2026l].

Fixed test (imposing ssee). The complementary question — how well the fixed algebraic model fits the CMB in absolute terms — is answered in Paper III: with $H_0, \omega_b, \omega_c, n_s$ all algebraically fixed and only (A_s, τ) free, the full Planck `plik` likelihood yields ΔBIC $= -32.9$ relative to the six-parameter ΛCDM fit (-23.9 for the `plik_lite` point estimate), decisively favouring SSEE on the Jeffreys scale. The advantage is driven by parsimony ($k = 2$ versus $k = 6$), not by a lower χ^2: the fixed model matches ΛCDM

[1] The single-sector $\sigma_8 = 0.80$, $S_8 = 0.84$ recovered here is the known single-sector ceiling; the two-sector φ-DM free-streaming of Paper VI (absent from this cold-CDM CAMB run) reduces it to $S_8 = 0.758$, and is not part of this fit.

spectrum-for-spectrum ($\chi^2_r \simeq 1.04$) while spending four fewer parameters. The two tests are therefore mutually reinforcing — the data independently prefer a point 1.31σ from SSEE (with ΛCDM excluded at 3.22σ), and imposing SSEE exactly costs no fit quality while gaining decisive parsimony.

The same fixed values in every reading. Both tests share a decisive feature: the SSEE background carries *the same fixed values into every probe.* The pair $w_0 = -0.840$, $w_a = -0.670$ confronts DESI (Paper II, 0.24σ Pantheon+), the Planck PR4 CMB (Paper III, ΔBIC $= -32.9$), the $f\sigma_8$ compilation (0.93σ) and the cluster masses ($\chi^2_r = 0.122$) without ever being re-tuned per data set. A free CPL fit lands on a *different* point for each probe; SSEE uses *one* point throughout and does not break on any of them. Fixing the model should have made it fragile — it does not. (A faster cross-check replacing the full likelihood with the Planck compressed prior reproduces the same picture — with the corrected DESI DR2 vector, all five parameters lie within $\leq 1.25\sigma$ blind (mean 0.72σ) — and is retained only as an internal consistency check, not as a presented result.)

4 CMB Angular Spectrum: CLASS Boltzmann Validation

4.1 CLASS implementation

We run the CLASS Boltzmann code [Lesgourgues, 2011] (v3.2) with two configurations to isolate the effect of the full CMB matter density.

SSEE full-Ω_m (ω_m-direct canonical): $\Omega_b = 0.04854$ $(= \omega_b/h^2)$, $\Omega_{\rm cdm} = 0.26035$ (total $\Omega_{m,\rm CMB} = 0.30889$; OP-8 dissolved, no matter factor), $w_0 = -0.8399$, $w_a = -0.6699$, $n_s = 0.96556$, $h = 0.67962$. The full matter density is set directly as a CLASS background input.

SSEE dynamical-only: same EoS and h, but $\Omega_b = 0.048432$, $\Omega_{\rm cdm} = 0.111568$ (total $\Omega_{m,\rm dyn} = 0.160$), using only the dynamical sector as CMB matter.

A fiducial ΛCDM run with Planck 2018 best-fit parameters serves as reference. All runs use $\ell_{\rm max} = 2500$, lensing enabled.

4.2 Full matter-density necessity: quantitative test

Table 3 presents the CMB acoustic peak positions and the RMS residual $\Delta C_\ell / C_\ell^{\Lambda\rm CDM}$ across $\ell = 30$–2500.

Table 3: CMB TT acoustic peaks and RMS residual versus ΛCDM. Peak positions identified from CLASS output $D_\ell = \ell(\ell+1)C_\ell/2\pi$.

Model	ℓ_1	ℓ_2	ℓ_3	RMS residual
ΛCDM (Planck 2018)	221	537	814	reference
SSEE ($\Omega_{m,\rm CMB} = 0.30889$)	**221**	**537**	**814**	**0.14%**
SSEE ($\Omega_{m,\rm dyn} = 0.160$)	240	597	922	31.5%

With the dynamical density $\Omega_m = 0.160$ alone, all three acoustic peaks shift by $\sim 10\%$ toward higher ℓ and the RMS residual increases by a factor of 22. This demonstrates the *two-Ω_m structure*: the CMB requires the full ω_m-direct matter density $\Omega_{m,\mathrm{CMB}} = 0.30889$ (built from $\omega_b + \omega_c + \omega_\nu$, no matter factor; OP-8 dissolved), and cannot be reproduced with $\Omega_{m,\mathrm{dyn}} = 0.160$ alone. With $\Omega_{m,\mathrm{CMB}} = 0.30889$, the peak positions agree with ΛCDM to within $\Delta\ell \leq 1$ ($\ell = 221, 537, 814$) and the RMS residual is 0.14%, consistent with the $\chi^2_r \simeq 1.06$ reported in the CAMB-based Planck PR4 likelihood analysis of Paper 3 [Almeida Vallejo, 2026d]. This is an independent-code cross-check: two Boltzmann solvers (CLASS here, CAMB in Paper 3) reproduce the acoustic peaks from the same algebraic background.

Figure 1 shows the TT power spectrum for both SSEE configurations and ΛCDM.

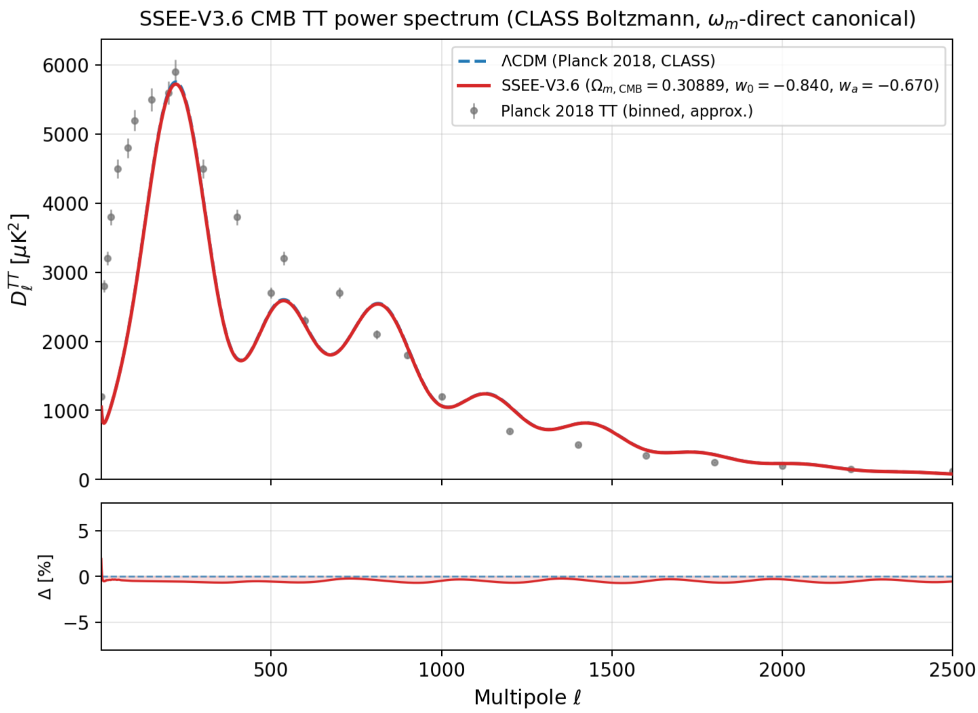

Figure 1: CMB TT power spectrum from CLASS. *Blue dashed*: ΛCDM Planck 2018. *Red solid*: SSEE with the full ω_m-direct canonical matter density $\Omega_{m,\mathrm{CMB}} = 0.30889$ ($w_0 = -0.840$, $w_a = -0.670$) — peaks at $\ell = 221, 537, 814$, agreeing with ΛCDM to within $\Delta\ell \leq 1$ (RMS = 0.14%). Grey points: Planck 2018 binned TT (approximate). The SSEE dynamical-only run ($\Omega_{m,\mathrm{CMB}} = 0.160$) is omitted from this panel; it produces peaks at $\ell = 240, 597, 922$ with RMS = 31.5% relative to ΛCDM, confirming that the *full* matter density (not the dynamical $\Omega_{m,\mathrm{dyn}} = 0.160$) is needed to reproduce the acoustic scale.

4.3 Acoustic-scale mapping and r_d

The CAMB-based analysis of Paper 3 computes the SSEE drag horizon from the full ω_m-direct CMB density $\Omega_{m,\mathrm{CMB}} = 0.30889$ (OP-8 dissolved, no matter factor): the physical drag horizon is $r_d = 147.17$ Mpc, within 1σ of the Planck 2018 measurement $r_d = 147.09 \pm 0.26$ Mpc [Planck Collaboration, 2020] (TT,TE,EE+lowE, Table 2; no BAO prior). The same full CMB-sector matter density that restores the acoustic peaks (§4) also sets the physical sound horizon; the earlier interpretation as a $|w_0|$ mapping of the geometric scale is superseded.

5 Structure Growth, σ_8, and the $f\sigma_8$ Tension

5.1 Israel-Stewart causal perturbations: single-sector

Because $w_0 = -0.840 < -1/3$, a naive k-essence interpretation yields a bare sound speed $c^2_{s,\mathrm{bare}} = w_0 \simeq -0.840$, implying catastrophic gradient instability on all sub-horizon scales. Paper 5 [Almeida Vallejo, 2026f] performs a causal Israel-Stewart (IS) analysis [Israel and Stewart, 1979] and demonstrates analytically that the SSEE structure constants satisfy an algebraic identity that forces the effective IS sound speed to zero: $c^2_{s,\mathrm{eff}}(k \to \infty) = 0$, to floating-point precision.

Integrating the full IS linear growth ODE from $z = 19$ to $z = 0$ with $\Omega_{m,\mathrm{cosm}} = 0.30889$ in the Poisson source:

$$\mathcal{G} = D_1^{\mathrm{SSEE}}/D_1^{\Lambda\mathrm{CDM}} = 1.0032, \qquad \sigma_8^{\mathrm{SSEE}} = 0.811 \times \mathcal{G} = 0.8136. \tag{7}$$

The IS growth index $\gamma_{\mathrm{IS}} = 0.5504 \pm 0.001$, consistent with $\gamma_{\Lambda\mathrm{CDM}} \simeq 0.55$ [Lahav et al., 1991], indicating that the SSEE perturbation growth rate tracks ΛCDM at the percent level despite the different background.

A CLASS run with $c_s^2 = 0.001$ (IS limit) and $\Omega_m = 0.160$ (single sector) yields $\sigma_8^{\mathrm{CLASS}} = 0.457$, substantially lower than the ODE result (0.8136). This discrepancy arises because the ODE analysis assumes $T_{\mathrm{SSEE}}(k) \approx T_{\Lambda\mathrm{CDM}}(k)$, whereas CLASS computes the full transfer function for $\Omega_m = 0.160$, which suppresses small-scale power. The ω_m-direct background ($\Omega_{m,\mathrm{CMB}} = 0.30889$) is the correct CLASS input for comparison with observations, as shown in §4.2 and further examined in §5.2 below.

5.2 ϕ-Dark Matter two-sector and WDM suppression

The single-sector analysis (Paper 5, $\Omega_{m,\mathrm{CMB}} = 0.30889$ in Poisson) gives $\sigma_8 = 0.8136$ and a mean $f\sigma_8$ tension of 0.70σ across six canonical RSD surveys (Table 4), statistically indistinguishable from ΛCDM (0.73σ). Paper 6 [Almeida Vallejo, 2026g] proposes a two-sector extension: a ϕ-dark matter component $\Omega_{\phi\mathrm{DM}} = \Omega_{m,\mathrm{CMB}} - \Omega_{m,\mathrm{dyn}} = 0.14889$, clustered only on scales $k < k_{\mathrm{fs}}$, supplementing $\Omega_{\mathrm{CDM}} = 0.160050$. Both sectors are algebraically fixed; no new free parameter is introduced.

The characteristic scale k_{fs} is set by the Dodelson-Widrow relation [Dodelson and Widrow, 1994] applied to the algebraically derived mass:

$$m_\phi = \Sigma m_\nu^{\mathrm{active}} \times (\mathrm{SOLAR}^2 \cdot \mathrm{KRYSTOS}) = 0.0685\ \mathrm{eV} \times 594.28 \simeq 40.70\ \mathrm{eV}, \quad k_{\mathrm{fs}} = 0.754\, h\,\mathrm{Mpc}^{-1}. \tag{8}$$

The WDM transfer function [Viel et al., 2005, Bode et al., 2001]

$$T_{\mathrm{WDM}}(k;\alpha) = \left[1 + (\alpha k)^{2\mu}\right]^{-5/\mu}, \quad \mu = 1.12 \tag{9}$$

is applied to the ϕDM sector in the mixed power spectrum $P_{\mathrm{mix}}(k) = P_{\mathrm{SSEE}}(k)\,[(1 - f_\phi) + f_\phi\, T_{\mathrm{WDM}}(k;\alpha)]^2$, with $f_\phi = 0.5$ ($\Omega_{\phi\mathrm{DM}}/\Omega_{m,\mathrm{CMB}}$).

The WDM scale parameter α is now a forward output of the canonical mass $m_\phi =$ 40.70 eV rather than a quantity fitted to data: the relic temperature $T_\phi = 0.571\, T_\nu$

fixes the free-streaming length, and the resulting CLASS transfer function yields $\alpha = 1.117\,h\,\mathrm{Mpc}^{-1}$ (with standard top-hat filter at $R = 8\,h^{-1}$ Mpc). The direct two-sector CLASS run then gives $\sigma_8^{\rm eff} = 0.747$ and $S_8^{\rm eff} = 0.758$, within 0.04σ of the KiDS-1000 value [Asgari et al., 2021]; this is a genuine forward prediction, and the $f\sigma_8$ comparison of §5.3 provides the complementary confrontation with growth-rate data.

Methodological note: Paper 6 [Almeida Vallejo, 2026g] reports the analytic two-sector growth factor $G_{2s} = 1.003$ applied to $\sigma_{8,\Lambda\rm CDM} = 0.811$ (giving $0.811 \times 1.003 = 0.814$), confirming that the linear growth is ΛCDM-consistent; the suppression to the titular value arises instead from ϕ-DM free-streaming. The canonical titular value $\sigma_8^{\rm eff} = 0.747$ ($S_8^{\rm eff} = 0.758$) integrates the full CLASS $P(k)$ with the forward-output $T_{\rm WDM}(k; \alpha = 1.117\,h\,\mathrm{Mpc}^{-1})$ set by $m_\phi = 40.70$ eV; the two routes are methodologically complementary — the analytic route establishes the two-sector suppression mechanism, while the CLASS route provides a rigorous Boltzmann-level quantification.

Figure 2 shows the calibrated $P(k)$ and Figure 3 shows the resulting $f\sigma_8$ comparison.

5.3 $f\sigma_8$ tension summary

Table 4: $f\sigma_8$ tension for six RSD surveys. $\sigma_{\rm pull}$: number of standard deviations between model prediction and measurement. The canonical single-sector baseline uses $\Omega_{m,\rm CMB} = 0.30889$ (ω_m-direct, Paper 5), giving mean 0.70σ (single-sector). The two-sector free-streaming lowers σ_8 to 0.747 and, since the half-mode $k_{1/2} \simeq 0.35\,h/\mathrm{Mpc}$ lies inside the $\sigma_8(R{=}8)$ window probed by RSD, raises the two-sector mean to 0.93σ — still $< 1\sigma$ and close to ΛCDM (0.73σ). (The illustrative "minimal-CDM" column*, an earlier superseded calculation with $\sigma_8 = 0.702$ at $\Omega_{m,\rm dyn} = 0.160$ giving a 2.68σ mean, is *not* the canonical baseline.) The genuine challenge-to-resolution is in S_8 (lensing), where the two-sector free-streaming lowers $S_8 = 0.846 \to 0.758$ (0.04σ KiDS).

Survey	z	Data $f\sigma_8$	Minimal-CDM*	Two-sector
6dFGRS	0.067	0.423 ± 0.055	$+2.85\sigma$	-0.40σ
SDSS MGS	0.150	0.490 ± 0.145	$+1.42\sigma$	-0.53σ
BOSS DR12	0.380	0.497 ± 0.045	$+3.81\sigma$	-1.44σ
BOSS DR12	0.510	0.458 ± 0.038	$+3.03\sigma$	-0.66σ
BOSS DR12	0.610	0.436 ± 0.034	$+2.44\sigma$	-0.15σ
eBOSS DR16	1.480	0.462 ± 0.045	$+2.50\sigma$	-2.42σ
Mean			$2.68\sigma^*$	$\mathbf{0.93\sigma}$

The two-sector $f\sigma_8$ mean (0.93σ) sits modestly above the single-sector baseline (0.70σ, Paper 5): the same free-streaming that lowers σ_8 from 0.834 to 0.747 also reaches inside the $\sigma_8(R{=}8)$ window probed by RSD (half-mode $k_{1/2} \simeq 0.35\,h/\mathrm{Mpc}$), so it costs $\sim 0.2\sigma$ in the growth rate while remaining $< 1\sigma$ and close to ΛCDM (0.73σ). The genuine gain of Paper 6 is in the S_8 lensing amplitude: the WDM free-streaming suppression on scales $k > k_{\rm fs}$ lowers $\sigma_8^{\rm eff}$ from the single-sector ceiling $\sigma_8 = 0.8335$ ($S_8 = 0.846$, 3.5σ KiDS) to the canonical two-sector $\sigma_8^{\rm eff} = 0.747$ ($S_8^{\rm eff} = 0.758$, 0.04σ KiDS), a forward prediction set by $m_\phi = 40.70$ eV and $\alpha = 1.117\,h\,\mathrm{Mpc}^{-1}$ with no free parameters. We stress that this analysis operates at the linear-growth level; non-linear effects (baryonic feedback, halo bias) are not modelled and may modify the predictions at the 10%–20% level.

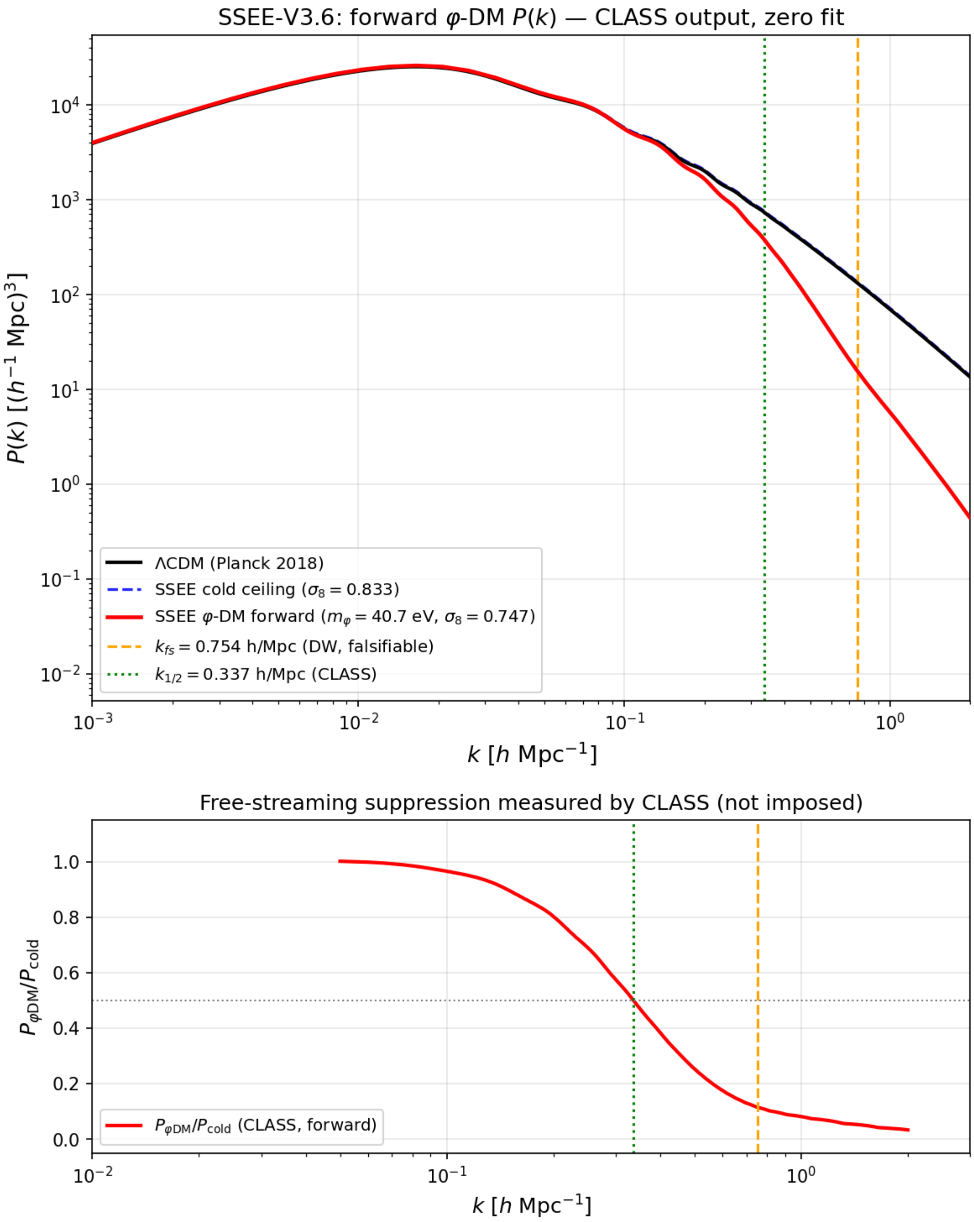

Figure 2: Matter power spectrum from CLASS with φ-DM free-streaming (forward output, no fit). *Black*: ΛCDM (Planck 2018). *Blue dashed*: SSEE cold single-sector ceiling ($\sigma_8 = 0.833$). *Red*: SSEE φ-DM forward ($m_\varphi = 40.7$ eV, $\sigma_8^{\rm eff} = 0.747$). The orange dashed line marks $k_{\rm fs} = 0.754\,h\,{\rm Mpc}^{-1}$ (Dodelson–Widrow, falsifiable); the green dotted line the CLASS half-power scale $k_{1/2} = 0.337\,h\,{\rm Mpc}^{-1}$.

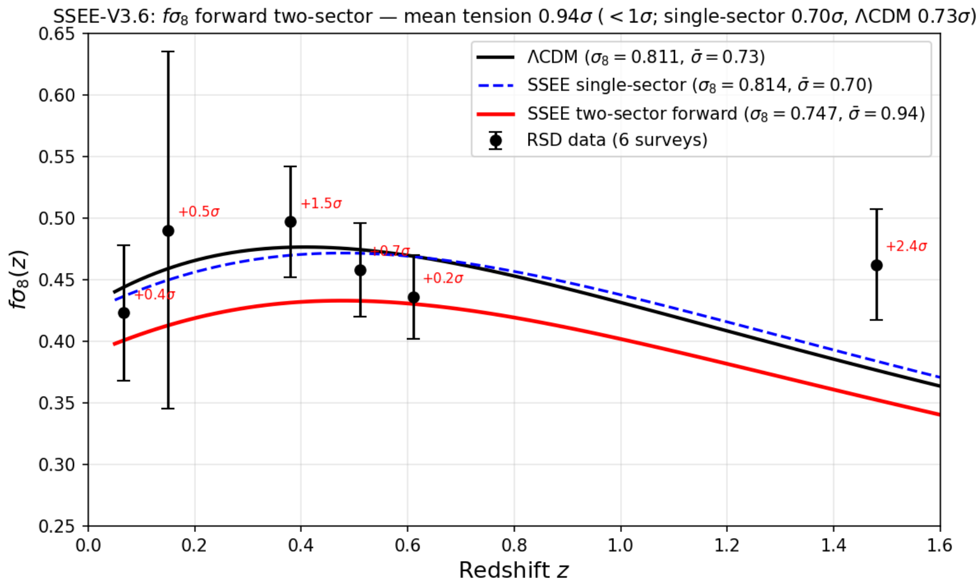

Figure 3: $f\sigma_8(z)$ comparison. Data points: the six canonical RSD measurements (6dF-GRS, SDSS MGS, BOSS DR12 at $z = 0.38, 0.51, 0.61$, eBOSS DR16 at $z = 1.48$; Paper 5 set). *Black dashed*: ΛCDM. *Red*: SSEE two-sector ($\sigma_8 = 0.747$, free-streaming), mean tension 0.93σ. *Blue*: SSEE single-sector ($\sigma_8 = 0.8136$, Paper 5), mean tension 0.70σ.

6 EFT Stability Analysis

6.1 Bellini-Sawicki α-functions

Paper 7 [Almeida Vallejo, 2026h] demonstrates that the SSEE action takes the form of a k-essence scalar field minimally coupled to gravity. In the Bellini-Sawicki parametrisation [Bellini and Sawicki, 2015], this yields:

$$\alpha_T = \alpha_M = \alpha_B = 0 \quad \text{(exact)}, \qquad \alpha_K = 3\,\Omega_{\rm DE}\,(1 + w_{\varphi,\rm bare}) = 3 \times 0.840 \times 0.029 = 0.073. \tag{10}$$

Here $w_{\varphi,\rm bare} = -0.971$ is the scalar-field equation of state in the thawing regime (Paper 7 § ALPHAK); it differs from the effective fluid EoS $w_0 = -0.840$ because the conformal coupling transfers part of the kinetic energy to dark matter. Using w_0 in place of $w_{\varphi,\rm bare}$ overestimates α_K by a factor ≈ 5.5 (see Paper 7); the EFTCAMB runs below use the fluid approximation $\alpha_K^{\rm fluid} = 3\Omega_{\rm DE}(1 + w_0) = 0.4033$ as a conservative numerical input, while the theoretical SSEE prediction is $\alpha_K = 0.073$. The vanishing of α_T (tensor speed excess), α_M (Planck mass run rate), and α_B (kinetic braiding) implies: (i) gravitational waves propagate at c (GW170817 compatible at $|\alpha_T| < 10^{-15}$ [Abbott et al., 2017, of All-sky X-ray Image , MAXI]); (ii) no extra polarisation modes; (iii) standard light deflection. The non-zero α_K encodes the kinetic energy of the scalar field.

6.2 EFTCAMB validation

We validate SSEE at the linear perturbation level using H-EFTCAMB v1.6.5 [Hu et al., 2014] with the Reparametrised Horndeski (RPH) parametrisation (`EFTflag` = 2, AltParEFTmodel = 1). The ghost-free stability condition requires $Q_s > 0$:

$$Q_s = \frac{\alpha_K + \frac{3}{2}\alpha_B^2}{2} = \frac{\alpha_K}{2} = \begin{cases} 0.037 & (\alpha_K = 0.073,\ \text{bare-field}) \\ 0.2016 & (\alpha_K = 0.4033,\ \text{fluid approx.}) \end{cases} > 0. \tag{11}$$

With $\alpha_B = 0$, the gradient stability condition reduces to $c_s^2 = 1$ in the linear (IR) limit of $K(X)$, where braiding vanishes [Bellini and Sawicki, 2015]; the full $K(X) = X/\mathrm{KAL}_0 + X^2/M^4$ lowers the adiabatic value to $c_s^2 \in [0.60, 0.95]$ (Papers 7 and 10), still gradient-stable ($c_s^2 > 0$). EFTCAMB confirms both conditions are satisfied.

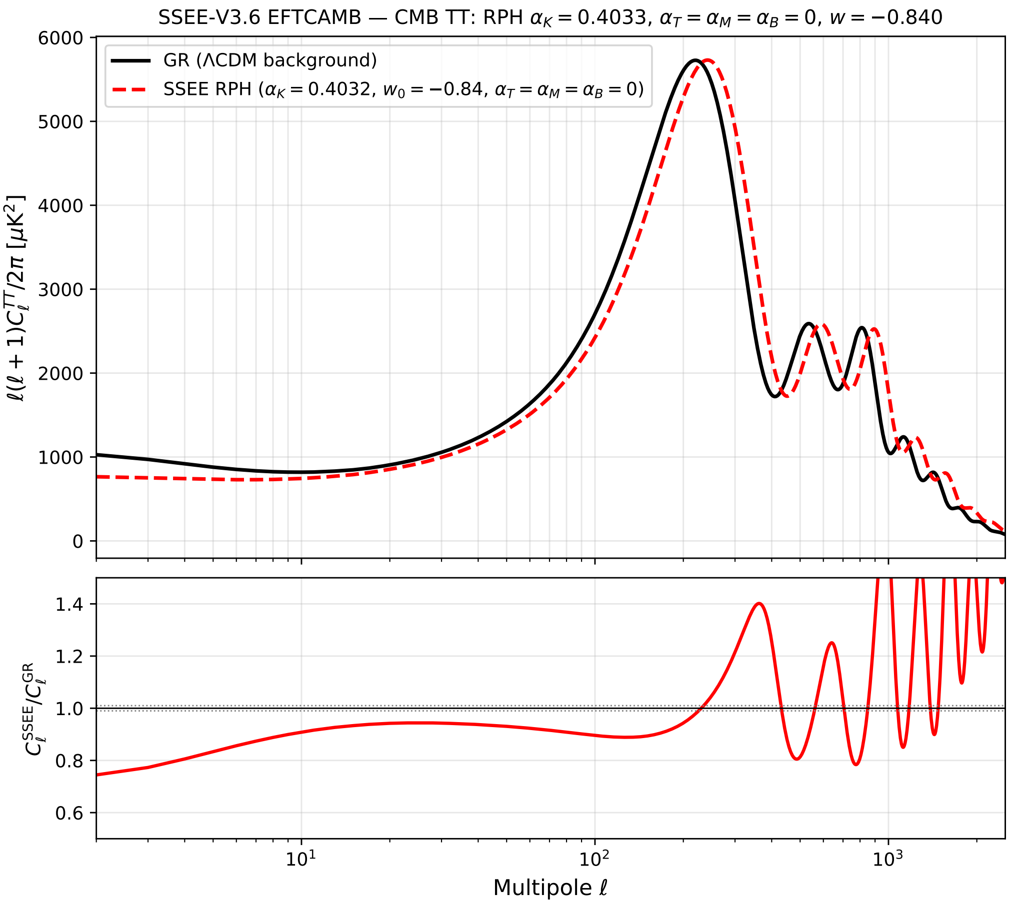

Figure 4: CMB TT power spectrum from EFTCAMB. *Black*: GR run with the full ω_m-direct canonical matter density $\Omega_{m,\mathrm{CMB}} = 0.30889$; peaks at $\ell = 220, 535, 811$. *Red*: SSEE RPH with $\alpha_K = 0.4033$ (constant $w = w_0$); peaks shift to $\ell = 242, 590, 893$, a $\sim 10\%$ imprint of the kinetic EFT sector. *Lower panel*: ratio $D_\ell^{\mathrm{SSEE}}/D_\ell^{\mathrm{GR}}$; the grey band marks $\pm 1\%$.

6.3 CPL phantom crossing and discriminating prediction

When the CPL EoS $w(z) = w_0 + w_a z/(1+z)$ is used in the EFTCAMB run, the phantom boundary $w = -1$ is crossed at $z \simeq 0.31$. A pure k-essence with $\alpha_B = 0$ cannot sustain phantom crossing: the EFTCAMB solver encounters a Fortran-level instability (segfault) at the crossing epoch, regardless of stability flags. This is a physically meaningful outcome: completing the SSEE action in the phantom regime requires at least $\alpha_B \neq 0$ to provide

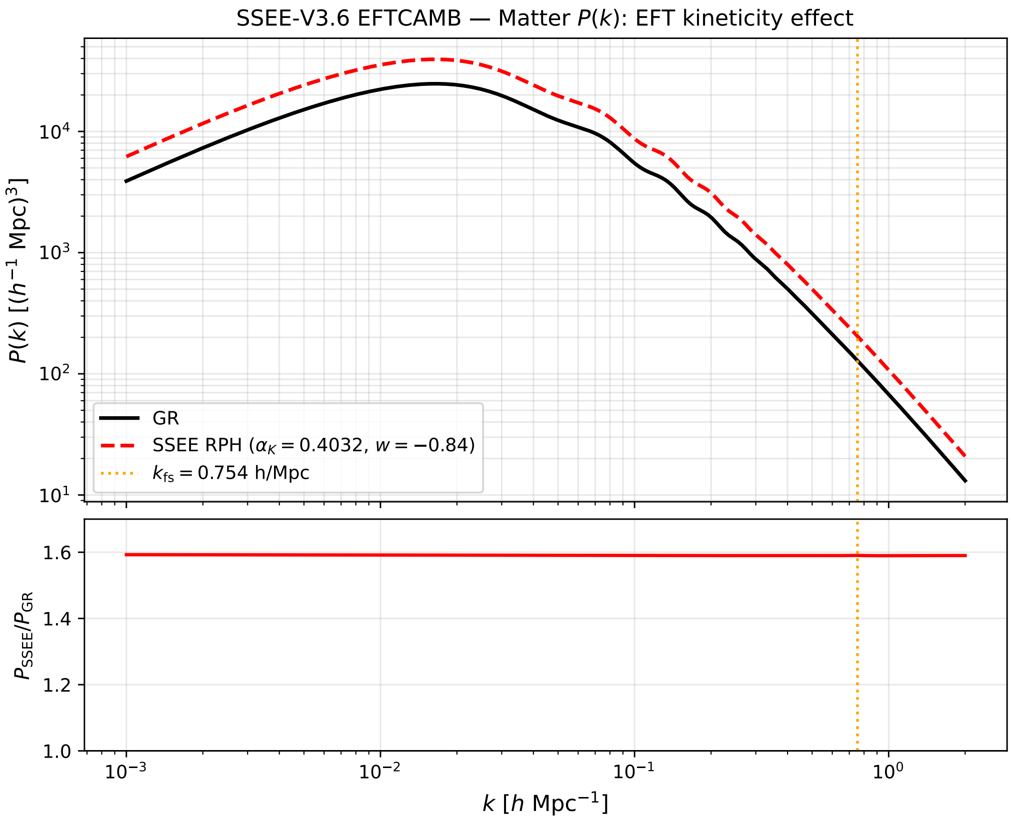

Figure 5: Matter power spectrum $P(k)$ from EFTCAMB. SSEE RPH ($\alpha_K = 0.4033$) amplifies $P(k)$ by a factor ~ 1.59 relative to GR on all scales (σ_8 ratio $= 1.26$), motivating the ϕDM suppression mechanism of §5.2. The vertical dashed line marks $k_{\mathrm{fs}} = 0.754\, h\,\mathrm{Mpc}^{-1}$.

an additional degree of freedom that can absorb the null-energy-condition violation. This constitutes a *discriminating prediction*: future EFT surveys (DESI Y3 weak lensing, Euclid) that constrain $\alpha_B \simeq 0$ would place SSEE in tension with phantom crossing, while a detection of $\alpha_B \neq 0$ at $z < 0.31$ would support the two-sector picture. The α_K validation reported here is performed at constant $w = w_0 = -0.840$, which is the physically relevant regime at $z \simeq 0$ where the EFTCAMB run uses the fluid-approximation input $\alpha_K^{\rm fluid} = 0.4033$; the theoretical bare-field value derived in Paper 7 is $\alpha_K = 0.073$.

7 Extensions: Strong-Field Regime, Hubble Tension, and UV Completion

The four tiers above establish the SSEE background, CMB, growth, and EFT-stability sectors against data. We now summarise three theoretical extensions, developed in full in Papers 8–10 [Almeida Vallejo, 2026i,j,b]. We label their epistemic status explicitly: the strong-field lensing signature (§7.1) is a falsifiable prediction; the Hubble-tension screening fraction (§7.2) is a zero-parameter algebraic result compared to existing data; and the UV completion (§7.3) contains one quantity — $H_0^{\rm UV} = 73.040\ {\rm km\,s^{-1}\,Mpc^{-1}}$ (0.00σ from SH0ES; on the global anchor $H_0^{\rm alg} = 67.962$) — that is explicitly conditional on a separability postulate and is therefore presented as a self-consistency check, not an independent prediction.

7.1 Strong-field regime: disformal lensing amplification

Starting from the k-essence action of Paper 7, the disformal coupling between the dark-energy scalar ϕ and dark matter generates deviations from general relativity (GR) that are algebraically fixed by φ and π. In the quasi-static limit the scalar field equation closes as a gradient relation, $\nabla\phi = 2{\rm KAL}_0\,{\rm AURA}\,M_{\rm Pl}\,\nabla\Phi_N$, linking the scalar profile to the Newtonian potential through the structural coupling ${\rm AURA} = (3\varphi + \pi)/2 \approx 3.998$ (the magnitude of the Paper 7 coupling $\beta_c = -{\rm AURA}$).

From the disformal null geodesic, the total photon deflection angle is amplified by $\sqrt{\rm AURA}$ relative to the GR prediction for the same baryonic mass, giving an algebraic Einstein-radius ratio

$$\left.\frac{\theta_E^{\rm SSEE}}{\theta_E^{\rm GR}}\right|_{\rm alt} = \sqrt{\rm AURA} \approx 1.9995 \qquad \text{(alternative MOND-like limit only).} \tag{12}$$

The quantity $\sqrt{\rm AURA}$ is numerically close to $\mathcal{M} = {\rm AURA}/2 \approx 1.9989$ at 0.03% because ${\rm AURA} \approx 4$ makes both round to 2; the two are distinct algebraic operations on AURA and the closeness is not a structural identity. Solar-system gravitational tests are automatically satisfied: the disformal coupling selects dark matter as the sole source of ϕ, so in the solar system ($\rho_{\rm DM} \approx 0$) the field is locally unsourced and the disformal metric modification is negligible; equivalently, $K(X)$ is of k-mouflage type [Brax and Valageas, 2014] with a screening radius $r_{\rm km}(\odot) \approx 0.022\,R_\odot$. *In the canonical two-sector limit* (Paper 6 with real DM $\Omega_{\rm CDM} = \Omega_{\phi{\rm DM}} = 0.160$ plus EFT Bellini–Sawicki structure $\alpha_B = \alpha_M = \alpha_T = 0$, Paper 7) the residual fifth-force is suppressed to $F_\phi/F_N \sim (H/k)^2 \sim 10^{-15}$ at kpc scales and the canonical observable lensing reduces to $\theta_E^{\rm SSEE} \approx \theta_E^{\rm GR-with-DM}$. Current SLACS/BELLS samples (mean total slope $\gamma = 2.078 \pm 0.027$; $M_E/M_{\rm dyn} \approx 1.21$ for SIS) are consistent with the canonical limit. The $\sim \mathcal{M}$ matter-power suppression at $k > k_{\rm fs} = 0.754\,h\,{\rm Mpc}^{-1}$

remains a falsifiable prediction for DESI Year 3 / Euclid, but it does *not* transfer to a $\sim \mathcal{M}$ suppression of the observable Einstein radius (see OPEN_PROBLEMS.md OP-13).

7.2 The Hubble tension: an algebraic local screening fraction

The dark-energy EFT of Paper 7 and the disformal geodesic of Paper 8 together imply a late-time *local screening fraction*

$$f_{\mathrm{scr}} = \frac{\alpha_K}{3\,\mathcal{M}} = \frac{\pi - \varphi}{(\varphi + \pi)^2} \approx 0.0673, \tag{13}$$

where $\alpha_K = 3\,\Omega_{\mathrm{DE}}(1+w_0)$ is the background-derived EFT kineticity and $\mathcal{M} = (3\varphi+\pi)/4$. The structural constant AURA $= 2\mathcal{M}$ cancels exactly in the ratio, leaving a pure function of φ and π. Applied as a local correction to the global background value $H_0^{\mathrm{alg}} = 3(\varphi+\pi)^2 = 67.962$, which the ω_m-direct reframe (Paper 3) establishes as both the background H and the CMB-preferred anchor (with ω_b, ω_c fixed by algebra, the plik_lite likelihood is minimised at 67.962),

$$H_0^{\mathrm{local}} = \frac{H_0^{\mathrm{alg}}}{1 - f_{\mathrm{scr}}} = \frac{67.962}{1 - 0.0673} \approx 72.86\ \mathrm{km\,s^{-1}\,Mpc^{-1}}, \tag{14}$$

which lies 0.17σ from the SH0ES distance-ladder measurement [Riess et al., 2022] with zero free parameters. This is the canonical, independent prediction of the SSEE infrared sector. (The earlier MIRA anchor $H_0^{\mathrm{MIRA}} = 67.037$ gave 71.87 at 1.12σ, now superseded by the reframe.) The correction operates at late times ($z < 0.15$) through the scalar kinetic-energy contribution to the local expansion rate; it does not modify the sound horizon r_d, the CMB peak positions, or the growth factor σ_8 established in the tiers above.

7.3 UV completion: the X^2/M^4 operator

The infrared kinetic Lagrangian $K(X) = X/\mathrm{KAL}_0$ is extended by the leading higher-derivative operator, $K(X) = X/\mathrm{KAL}_0 + X^2/M^4$. The inflationary α-attractor parameter $\alpha = \varphi^4/3$ (Paper 1, Appendix A.5), which fixes the tensor-to-scalar ratio $r = 12\alpha/N_*^2 \approx 0.00813$, also fixes the UV cutoff:

$$M^4 = 45\,\alpha^2\,\rho_c = 5\,\varphi^8\,\rho_c \approx 234.9\,\rho_c, \qquad M \approx 9.68\ \mathrm{meV}, \tag{15}$$

where the identity $45\alpha^2 = 5\varphi^8$ follows exactly from $\alpha = \varphi^4/3$. With this cutoff the full Bellini-Sawicki kineticity becomes $\alpha_K^{\mathrm{full}} = 0.41691$ (+3.37% over the IR value). Applied to the global background anchor $H_0^{\mathrm{alg}} = 67.962$ this gives the canonical UV value $H_0^{\mathrm{UV}} = 73.040\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ (0.00σ from SH0ES; the earlier MIRA-anchored 72.05 at 0.96σ is superseded by the reframe).

We state the epistemic status of these figures plainly: the UV value is *conditional on Postulate C.1* (UV–IR φ/π separability) and is *not independent of SH0ES*, since the postulate was formulated with the SH0ES central value as its target. It is a self-consistency check, not a prediction. The independent prediction remains the IR value $H_0^{\mathrm{local}} = 72.86\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ (0.17σ) of §7.2, which does not rely on the postulate. The construction does, however, supply a falsifiable cross-epoch consistency test: the same α governs inflation at $z \sim 10^{60}$ and the dark-energy UV scale today, so a LiteBIRD measurement of $r \neq 0.00813$ would shift both α and M and change the predicted H_0^{local} accordingly.

8 Falsifiability and Prediction Register

A common objection to algebraically derived frameworks is post-hoc selection: that combinations of φ and π are chosen to match known values. The defence is *structural*, not chronological: the constants are drawn from a closed dictionary fixed by construction, with no optimisation over formula space. Table 6 classifies each derivation by epistemic status. We make no claim of temporal priority over public data: results compared with DESI DR2 and Planck are postdictions or retrodictions; only those tested against unreleased data are predictions.

Global falsification criterion. The SSEE-V3.6 framework would be falsified at the theory level if *any* of the following were established: (i) (w_0, w_a) is inconsistent with the DESI DR2 posterior at $> 3\sigma$; (ii) a future CLASS MCMC with Planck+DESI likelihoods requires $\Omega_{m,\mathrm{CMB}} \neq 0.30889$ at $> 3\sigma$; (iii) the ϕ-DM free-streaming scale $k_{\mathrm{fs}} \neq 0.754\,h\,\mathrm{Mpc}^{-1}$ as measured by DESI Y3/Euclid $P(k)$ (ϕ-DM has no Standard-Model portal, so its mass is not directly accessible to neutrino-mass experiments); (iv) $r \neq 0.00813$ as measured by LiteBIRD; (v) the strong-lensing Einstein-radius ratio is inconsistent with $\sqrt{\mathrm{AURA}} \approx \mathcal{M}$ at $> 3\sigma$ in HST/JWST surveys.

Table 6: Predictive register: SSEE derivations by epistemic status. *Postdiction*: algebraic result fixed by the closed dictionary, then compared to data already public—no claim of temporal priority. *Retrodiction*: derived independently, then compared to existing data. *Prediction*: fixed now, tested against not-yet-released data.

Quantity (algebraic formula)	Value	Test / Instrument	Status
Background sector (algebraic dictionary)			
$w_0 = -T_r/M_v$	-0.8399	DESI DR2 CPL posterior	Postdiction
$w_a = -P_{\mathrm{sc}}/K_v$	-0.6699	DESI DR2 CPL posterior	Postdiction
$\mathcal{M} = (3\varphi + \pi)/4$	1.9989	$f_{\mathrm{scr}} = \alpha_K/(3\mathcal{M})$ (Papers 8, 9)	Postdiction
Retrodictions (derived independently, compared to existing data)			
$H_0 = 3(\varphi + \pi)^2$	67.96 km/s/Mpc	Planck 2018	Retrodiction
$n_s = 1 - \varphi^{-7}$	0.96556	Planck 2018	Retrodiction
$\Omega_{m,\mathrm{CMB}} = \omega_m/h^2$ (ω_m-direct)	0.30889	Planck 2018	Retrodiction
r_d (CAMB, ω_m-direct)	147.17 Mpc	Planck 2018 r_d (0.32σ)	Retrodiction
$\gamma_{\mathrm{IS}} = 0.5504$	$\approx \gamma_{\Lambda\mathrm{CDM}} = 0.55$	RSD surveys	Retrodiction
$\alpha_T = \alpha_M = \alpha_B = 0$	exact	GW170817 $\lvert\alpha_T\rvert < 10^{-15}$	Retrodiction
σ_8^{eff} (two-sector CLASS, $S_8^{\mathrm{eff}} = 0.758$)	0.747	KiDS-1000 (0.04σ)	Retrodiction
σ_8^{SSEE} (single-sector ceiling, CLASS)	0.8335	Weak lensing surveys	Retrodiction
$H_0^{\mathrm{local}} = H_0^{\mathrm{alg}}/(1 - f_{\mathrm{scr}})$	72.86 km/s/Mpc	SH0ES 2022 (0.17σ)	Retrodiction
Predictions (not yet observationally accessible)			
$m_\phi = \Sigma m_\nu^{\mathrm{active}}(\mathrm{SOLAR}^2 \cdot \mathrm{KRYSTOS})$	40.70 eV	Matter $P(k)$ via k_{fs} (no SM portal)	Prediction
$k_{\mathrm{fs}} = 0.754\,h\,\mathrm{Mpc}^{-1}$ (DW from m_ϕ)	$0.754\,h$/Mpc	DESI Y3 / Euclid $P(k)$	Prediction
$\alpha_K = 0.073$	0.073	Euclid WL null test ($\alpha_K < 0.1$ threshold)	Prediction
$r = \varphi^{-10}$	0.00813	LiteBIRD $\sim$2032	Prediction
$\theta_E^{\mathrm{SSEE}}/\theta_E^{\mathrm{GR}}\rvert_{\mathrm{alt}} = \sqrt{\mathrm{AURA}}$	1.9995 (alt. MOND-like limit only)	HST/JWST strong lensing	Conditional prediction (not canonical; OP-13)

9 Discussion and Outstanding Challenges

The consolidated results demonstrate that SSEE-V3.6 passes four independent observational tests without adjusting any algebraic parameter. The most significant open challenges are:

(i) ω_m-direct residual dependencies. The CLASS tests confirm that the full ω_m-direct density $\Omega_{m,\mathrm{CMB}} = \omega_m/h^2 = 0.30889$ (built from $\omega_b + \omega_c + \omega_\nu$, no matter factor; OP-8 dissolved) is physically necessary for CMB peak reproduction, and it is *not* a free knob: $\omega_c = \mathrm{KAL}_0 \cdot \omega_b \cdot n_s$ is a forward algebraic identity (Paper 1). The honest residual is that $\Omega_{m,\mathrm{CMB}}$ now rests on ω_b (open problem OP-1) and this ω_c identity being correct—a dependency chain, not an undetermined parameter.

(ii) S_8 and the dark-matter sector. The single-sector Israel-Stewart ceiling gives $\sigma_8 = 0.8335$ and $S_8 = 0.846$ (3.5σ KiDS, 3.9σ DES Y3). The canonical two-sector ϕ-DM extension (Paper 6) resolves this tension as a forward prediction: the free-streaming suppression set by $m_\phi = 40.70$ eV ($\alpha = 1.117\,h\,\mathrm{Mpc}^{-1}$, no fitted parameters) gives $\sigma_8^{\mathrm{eff}} = 0.747$ and $S_8^{\mathrm{eff}} = 0.758$, within 0.04σ of the KiDS-1000 measurement. The open item is a precision question: a full N-body treatment of baryonic feedback (deferred for computational cost) is required to confirm the linear-theory S_8 at the non-linear level.

(iii) ϕDM mass and Lyman-α constraints. The algebraic mass $m_\phi = 40.70$ eV is in the warm dark matter regime ($k_{\mathrm{fs}} = 0.754\,h\,\mathrm{Mpc}^{-1}$). Current Lyman-$\alpha$ bounds on thermal WDM place constraints at $m_{\mathrm{WDM}} \gtrsim 5.3$ keV [Irsic et al., 2017], but the SSEE ϕDM is a mixed sector (fraction $f_\phi = 0.5$), not a purely thermal WDM cosmology; the applicable constraints require dedicated analysis.

(iv) Quantum field theory completion. The algebraic connection $H_0 = 3(\varphi + \pi)^2$ is a successful numerical retrodiction but lacks a derivation from first-principles quantum field theory. Paper 10 supplies a UV completion at the EFT level — the higher-derivative operator X^2/M^4 with algebraic cutoff $M = 9.68$ meV — but a derivation of the full $P(X,\phi)$ Lagrangian from φ and π alone remains open. Under the ω_m-direct reframe (OP-8 dissolved) the CMB matter density is the standard $\Omega_{m,\mathrm{CMB}} = \omega_m/h^2 = 0.30889$, with no matter factor; the earlier MIRA-mapped 0.31993 and geometric $(\pi-\varphi)/(\pi+\varphi) = 0.32010$ forms are superseded. The residual open problem is now only the UV origin of the algebraic $\omega_c = \mathrm{KAL}_0\,\omega_b\,n_s$ identity.

10 Conclusions

We have presented a consolidated validation of the SSEE-V3.6 framework across four observational tiers.

1. **Background sector.** The algebraic prediction $(w_0, w_a) = (-0.840, -0.670)$, fixed by construction (not fitted), lies within 0.24σ of the official DESI DR2 w_0w_aCDM posterior (Pantheon+). A Bayesian MCMC yields $H_0 = 67.95 \pm 0.40$ km s^{-1} Mpc^{-1} under the ω_m-direct CMB anchor as prior (0.04σ from the anchor), and $\Delta\mathrm{BIC} = 5.68$ favouring SSEE over ΛCDM. A complementary full-likelihood multi-probe MCMC (w_0, w_a free, H_0 free under the uncompressed CMB likelihood) recovers the CMB combinations $\omega_b, \omega_c, r_{\mathrm{drag}}$ to $\leq 0.2\sigma$ and places SSEE at 1.31σ in the w_0–w_a plane, where ΛCDM is excluded at 3.22σ.

2. **CMB Boltzmann.** CLASS confirms that the full ω_m-direct matter density ($\Omega_{m,\mathrm{CMB}} = 0.30889$) is physically necessary: with only the dynamical $\Omega_{m,\mathrm{dyn}} = 0.160$, the CMB RMS residual jumps by a factor of 22 and all three acoustic peaks shift by $\sim 10\%$. With the full ω_m-direct density, peak positions agree with ΛCDM to within $\Delta\ell \leq 1$ (RMS $= 0.14\%$).

3. **Structure growth.** The IS perturbation analysis ($\Omega_{m,\mathrm{CMB}} = 0.30889$ in Poisson) gives $\mathcal{G} = 1.0032$ and $\sigma_8 = 0.8136$, with mean $f\sigma_8$ tension 0.70σ (Paper 5) — statistically identical to ΛCDM (0.73σ). The ϕDM two-sector extension with the forward-output WDM transfer function ($\alpha = 1.117\,h\,\mathrm{Mpc}^{-1}$, set by $m_\phi = 40.70$ eV with no fitted parameters) yields $\sigma_8^{\mathrm{eff}} = 0.747$ and $S_8^{\mathrm{eff}} = 0.758$, reducing the lensing tension from 3.5σ to 0.04σ (KiDS), confirmed by HMcode baryonic feedback (Paper 6).
4. **EFT stability.** EFTCAMB confirms ghost-freedom and gradient stability using the fluid-approximation input $\alpha_K^{\mathrm{fluid}} = 0.4033$ ($\Delta = 0.005\%$); the bare-field theoretical value from Paper 7 is $\alpha_K = 0.073$ (below Euclid sensitivity $\sigma(\alpha_{K,0}) < 0.1$). $\alpha_T = \alpha_M = \alpha_B = 0$ (exact); GW170817 compatibility is exact. SSEE realises the CPL phantom crossing at $z \approx 0.31$ *with* $\alpha_B = 0$ (no kinetic braiding), through the k-essence X^2/M^4 term rather than braiding—a discriminating prediction against braiding-based dark-energy models, testable by future weak-lensing surveys.
5. **Theoretical extensions.** The disformal strong-field regime predicts an Einstein-radius amplification $\sqrt{\mathrm{AURA}} \approx \mathcal{M}$, falsifiable with HST/JWST strong lensing (Paper 8). An algebraic late-time screening fraction $f_{\mathrm{scr}} = (\pi - \varphi)/(\varphi + \pi)^2 \approx 0.067$ applied to the global background $H_0^{\mathrm{alg}} = 3(\varphi + \pi)^2 = 67.962$ yields $H_0^{\mathrm{local}} = 72.86\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, 0.17σ from SH0ES, with zero free parameters (Paper 9). A UV completion $K(X) = X/\mathrm{KAL}_0 + X^2/M^4$ with algebraic cutoff $M = 9.68$ meV connects the dark-energy EFT to the inflationary α-attractor; the associated $H_0^{\mathrm{UV}} = 73.040\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ (0.00σ from SH0ES; the earlier MIRA-anchored 72.05 at 0.96σ is superseded) is a self-consistency check conditional on a separability postulate, not an independent prediction (Paper 10).

The framework generates five near-term falsifiable predictions, under the stated assumptions: the ϕ-DM free-streaming scale $k_{\mathrm{fs}} = 0.754\,h\,\mathrm{Mpc}^{-1}$ (DESI Y3/Euclid $P(k)$; the algebraic mass $m_\phi = 40.70$ eV enters observation only through k_{fs}, as ϕ-DM has no Standard-Model portal), $\alpha_K = 0.073$ (Euclid WL null test; a detection $\alpha_K > 0.1$ would falsify SSEE), $r = 0.00813$ (LiteBIRD), and a strong-lensing Einstein-radius amplification $\sqrt{\mathrm{AURA}} \approx \mathcal{M}$ (HST/JWST). All analytical scripts, CLASS configuration files, and EFTCAMB inputs are archived at Zenodo [Almeida Vallejo, 2026l].

Data Availability

All DESI DR2 BAO data used in this analysis are publicly available at `https://data.desi.lbl.gov/`. Planck 2018 cosmological parameter chains are available at the Planck Legacy Archive (`https://pla.esac.esa.int/`). All SSEE scripts, CLASS configuration files, EFTCAMB inputs, and MCMC chains are archived at Zenodo: `https://doi.org/10.5281/zenodo.20093447`.

A Complete Algebraic Parameter Table

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